

# Mike Y. M. Lau

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Heidelberg Institute for Theoretical Studies (HITS)  
Centre for Astronomy (ZAH), Heidelberg University

## Timeline

- 09/23 – **Heidelberg Institute for Theoretical Studies**, Heidelberg, Germany  
Croucher Research Fellow (09/23–08/25)  
Postdoctoral researcher at Astronomisches Rechen-Institut, Centre for Astronomy of Heidelberg University (09/25–Present)
- 09/19 – 03/23 **Monash University**, Melbourne, Australia (PhD in Astrophysics)  
Dissertation title: *Interactions in Stellar Binaries*  
Supervisors: Ilya Mandel, Daniel J. Price, & Dr. Ryosuke Hirai
- 01 – 06/22 **Flatiron Institute Center for Computational Astrophysics**, New York, United States (Research Analyst in pre-doctoral program)  
Supervisors: Matteo Cantiello & Adam Jermyn
- 10/15 – 07/19 **University of Oxford**, Oxford, United Kingdom (MMathPhys)  
Master of Mathematical and Theoretical Physics with First Class in Parts A, B, & C  
Dissertation title: *Detecting Double Neutron Stars with LISA*  
Supervisors: Ilya Mandel & Philipp Podsiadlowski

## Grants & awards

- |         |                                                                                   |                                       |
|---------|-----------------------------------------------------------------------------------|---------------------------------------|
| 24      | Shortlist for Mollie Holman Doctoral Medal for Science                            | Monash University                     |
| 04/23   | Croucher Research Fellowship                                                      | Croucher Foundation                   |
| 03/23   | Humboldt Research Fellowship (declined for Croucher)                              | Alexander von Humboldt Foundation     |
| 01/23   | Postgraduate publication award                                                    | Monash University                     |
| 12/22   | Max Planck Institute for Astrophysics Fellowship (declined for Croucher)          | Max Planck Institute for Astrophysics |
| 21      | Lead CI for NCI Astronomy Program computing grant<br>Q1/2: 544 kSU, Q3/4: 670 kSU | AAL Astronomy Supercomputer           |
| 19 – 22 | J. L. William International PhD Scholarship                                       | Monash University                     |
| 19 – 23 | Research Training Program (RTP) Stipend                                           | Monash University                     |
| 19 – 23 | Monash International Tuition Scholarship                                          | Monash University                     |
| 07/19   | Schools Prize                                                                     | St Edmund Hall, University of Oxford  |
| 17, 18  | Open Scholarship                                                                  | St Edmund Hall, University of Oxford  |
| 16      | Open Exhibition                                                                   | St Edmund Hall, University of Oxford  |
| 08/15   | Hong Kong Scholarship for Excellence                                              | Hong Kong Government                  |

## Academic service & outreach

- Journal review *The Astrophysical Journal* (Letters and Main Journal), *Monthly Notices of the Royal Astronomical Society*, *Astronomy and Astrophysics*, & *The Open Journal of Astrophysics*
- Proposal review JWST Director's Discretionary Time, National Science Centre Poland
- 02/23 LOC, 2023 Phantom Users Workshop
- 11/22 OzGrav Outreach Superstar Award
- 10/19 – 08/23 Organiser, weekly Whiteboard Seminars at Monash University
- 09/22 World Science Festival, Ipswich, Queensland
- 07/21 Dark Science holiday programme, Casey Tech School (Berwick)
- 07/21 Black Hole Sunday, Twisted Science Centre, Echuca, Victoria
- 04/21 OzGrav Interactive tech showcase, Bendigo Discovery Science and Technology Centre
- 12/19 Monash Minimizer Faire, Monash University
- 18 Organiser for St Edmund Hall Physics Journal Club
- 08/17 – 19 Academic and Scholarship Mentor, Project Access HK

## Teaching & supervision

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| 24 –       | Ana Lourdes (Macquarie University), PhD co-supervision with Orsola De Marco                                                                      |
| 23 –       | Project co-supervision for PhD and Master’s students at the Heidelberg Institute for Theoretical Studies                                         |
| 11 – 12/21 | Co-supervisor for summer Honours student at Monash University                                                                                    |
| 02 – 06/21 | TA for <i>ASP3051 Relativity and Cosmology</i>                                                                                                   |
| 08 – 11/20 | Tutor for <i>ASP3162 Computational Astrophysics and the Extreme Universe</i> under the Monash University Indigenous Academic Enhancement Program |
| 08 – 11/20 | IAEP tutor for <i>ASP3012 Stars and Galaxies</i>                                                                                                 |
| 08 – 09/20 | Tutor for <i>MCD1180: Introductory Physics</i> under the Monash Indigenous Access Program                                                        |
| 04 – 06/20 | IAEP tutor for <i>ASP3051 Relativity and Cosmology</i>                                                                                           |
| 04 – 06/20 | IAEP tutor for <i>MAT9004 Mathematical Foundations for Data Science</i>                                                                          |
| 02 – 06/20 | TA for <i>ASP1010: Earth to Cosmos—Introductory Astronomy</i>                                                                                    |

## Publications

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*Published/accepted works (6 first-author, 2 second-author)*

21. Wang, C., **Lau, M.**, Li, X., Langer, N., de Mink, S. et al., 2026, *Thermal-timescale accretion does not always yield critical rotation in mass gainers*, [A&A](#), 707
20. Vetter, M., Röpke, F., Schneider, F., Pakmor, R., Ohlmann, S. et al. (inc. **Lau, M.**), 2025, *Magnetically driven outflows in the 3D common envelope evolution of massive stars*, [A&A](#), 698
19. Bermúdez-Bustamante, L., De Marco, O., Siess, L., Price, D., González-Bolívar, M. et al. (inc. **Lau, M.**), 2025, *Dust formation during the interaction of binary stars by common envelope*, *Planetary Nebulae (IAU S384) A Universal Toolbox in the Era of Precision Astrophysics*, [Proceedings of the IAU Symposia and Colloquia](#)
18. **Lau, M.**, Hirai, R., Price, D., Mandel, I., Bate, M., 2025, *Common envelopes in massive stars: III. The obstructive role of radiation transport in envelope ejection*, [A&A](#), 699
17. Mandel, I., Riley, J., Boesky, A., Brcek, A., Hirai, R. et al. (inc. **Lau, M.**), 2025, *Rapid stellar and binary population synthesis with COMPAS: methods paper II*, [ApJS](#), 280, 1
16. Vetter, M., Röpke, F., Schneider, F., Pakmor, R., Ohlmann, S. et al. (inc. **Lau, M.**), 2025, *Magnetically driven outflows in the 3D common envelope evolution of massive stars*, [A&A](#), 698
15. Siess, L., Bermúdez-Bustamante, L., De Marco, O., Price, D., González-Bolívar, M. et al. (inc. **Lau, M.**), 2024, *Dusty Common Envelope Evolution*, [Galaxies](#), 12
14. Schneider, F., **Lau, M.**, Röpke, F., 2025, *Stellar mergers and common-envelope evolution*, *Encyclopedia of Astrophysics - 1st Edition*, [Elsevier](#)
13. Vetter, M., Röpke, F., Schneider, F., Pakmor, R., Ohlmann, S. et al. (inc. **Lau, M.**), 2024, *From spherical stars to disk-like structures: 3D common-envelope evolution of massive binaries beyond inspiral*, [A&A](#), 691
12. Bermúdez-Bustamante, L., De Marco, O., Siess, L., Price, D., González-Bolívar, M. et al. (inc. **Lau, M.**), 2024, *Dust formation in common envelope binary interactions - II: 3D simulations with self-consistent dust formation*, [MNRAS](#), 533, 1
11. **Lau, M.**, Hirai, R., Mandel, I., Tout, C., 2024, *Expansion of Accreting Main-sequence Stars during Rapid Mass Transfer*, [ApJL](#), 966, 1
10. Amaro-Seoane, P., Andrews, J., Arca Sedda, M., Askar, A., Baghi, Q. et al. (inc. **Lau, M.**), 2023, *Astrophysics with the Laser Interferometer Space Antenna*, [Living Reviews in Relativity](#), 26, 1
9. Renzo, M., Zapartas, E., Justham, S., Breivik, K., **Lau, M.** et al., 2023, *Rejuvenated Accretors Have Less Bound Envelopes: Impact of Roche Lobe Overflow on Subsequent Common Envelope Events*, [ApJL](#), 942, 2
8. González-Bolívar, M., De Marco, O., **Lau, M.**, Hirai, R., Price, D., 2022, *Common envelope binary interaction simulations between a thermally pulsating AGB star and a low mass companion*, [MNRAS](#), 517, 3
7. **Lau, M.**, Hirai, R., Price, D., Mandel, I., 2022, *Common envelopes in massive stars II: The distinct roles of hydrogen and helium recombination*, [MNRAS](#), 516, 4
6. **Lau, M.**, Cantiello, M., Jermyn, A., MacLeod, M., Mandel, I. et al., 2025, *Hot Jupiter engulfment by an early red giant in 3D hydrodynamics*, [A&A](#), 694
5. **Lau, M.**, Hirai, R., González-Bolívar, M., Price, D., De Marco, O. et al., 2022, *Common envelopes in massive stars: towards the role of radiation pressure and recombination energy in ejecting red supergiant envelopes*, [MNRAS](#), 512, 4
4. Riley, J., Agrawal, P., Barrett, J., Boyett, K., Broekgaarden, F. et al. (inc. **Lau, M.**), 2022, *Rapid Stellar and Binary Population Synthesis with COMPAS*, [ApJS](#), 258, 2

3. Compas, T., Riley, J., Agrawal, P., Barrett, J., Boyett, K. et al. (inc. **Lau, M.**), 2022, *COMPAS: A rapid binary population synthesis suite*, *The Journal of Open Source Software*, 7, 69
2. Ackley, K., Adya, V., Agrawal, P., Altin, P., Ashton, G. et al. (inc. **Lau, M.**), 2020, *Neutron Star Extreme Matter Observatory: A kilohertz-band gravitational-wave detector in the global network*, *Publications of the Astronomical Society of Australia*, 37
1. **Lau, M.**, Mandel, I., Vigna-Gómez, A., Neijssel, C., Stevenson, S. et al., 2020, *Detecting double neutron stars with LISA*, *MNRAS*, 492, 3

#### Submitted works

24. **Lau, M.**, Andrassy, R., Leidi, G., Gagnier, D., Morán-Fraile, J. et al., 2026, *Continuous mass ablation of planets engulfed in stellar envelopes*, submitted to A&A, [arXiv:2601.08508](#)
23. Bernard, Y., David-Cléris, T., Price, D., **Lau, M.**, 2026, *Momentum-conserving self-gravity in the phantom smoothed particle hydrodynamics code. Parallel dual tree traversal for the symmetric fast multipole method*, submitted to A&A, [arXiv:2602.05804](#)
22. Heller, M., Schneider, F., Henneco, J., Bronner, V., **Lau, M.**, 2025, *1D stellar mergers: entropy sorting and PyMMAMS*, submitted, [arXiv:2512.14669](#)

#### Selected talks (\* = invited)

|       |                                                                                                                                      |                                        |
|-------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 12/25 | XIX Winter Workshop on Stellar Astrophysics 2025                                                                                     | <i>HITS, Heidelberg</i>                |
| 11/25 | Hong Kong Laureate Forum                                                                                                             | <i>Hong Kong</i>                       |
| 09/25 | *Astronomy seminar                                                                                                                   | <i>Tsung-Dao Lee Institute</i>         |
| 08/25 | *Binary Stars in a New Era                                                                                                           | <i>Lijiang</i>                         |
| 06/25 | *European Astronomical Society Annual Meeting<br>( <b>invited review</b> )                                                           | <i>Cork</i>                            |
| 06/25 | *2nd European Phantom code family users workshop                                                                                     | <i>IPAG, Grenoble</i>                  |
| 12/24 | *Astrophysics seminar                                                                                                                | <i>Chinese University of Hong Kong</i> |
| 07/24 | 41st Liège International Astrophysical Colloquium: The eventful life of massive star multiples ( <b>best linguistics invention</b> ) | <i>University of Liège</i>             |
| 02/24 | *Joint Franco-Australian 5th Phantom and MCFOST Users Workshop                                                                       | <i>Monash University, remote</i>       |
| 12/23 | *Astrophysics seminar                                                                                                                | <i>Chinese University of Hong Kong</i> |
| 03/23 | *Colloquium                                                                                                                          | <i>ICRAR-Curtin, Perth</i>             |
| 02/23 | Phantom users workshop 2023 (LOC)                                                                                                    | <i>Monash University</i>               |
| 01/23 | SESTAS meeting                                                                                                                       | <i>MPA, Garching</i>                   |
| 01/23 | Common envelope group meeting                                                                                                        | <i>HITS, Heidelberg</i>                |
| 12/22 | *Gravitational Wave Physics and Astronomy Workshop                                                                                   | <i>Melbourne</i>                       |
| 06/22 | CCA Predoctoral Program Symposium                                                                                                    | <i>CCA, Flatiron Institute</i>         |
| 06/22 | Physics and Astrophysics of Common Envelope                                                                                          | <i>Los Alamos National Laboratory</i>  |
| 03/22 | CCA Stars & Compact Objects Group Meeting                                                                                            | <i>CCA, Flatiron Institute</i>         |
| 12/21 | OzGrav Data/Astro Telecon                                                                                                            | <i>Virtual</i>                         |
| 09/21 | Common Envelope Physics and Outcomes 2021                                                                                            | <i>Virtual</i>                         |
| 07/21 | ASA Annual Meeting 2021                                                                                                              | <i>University of Melbourne</i>         |
| 07/21 | EAS Annual Meeting 2021                                                                                                              | <i>Leiden, virtual</i>                 |
| 02/21 | *LISA Workshop                                                                                                                       | <i>University of Auckland, remote</i>  |
| 08/20 | The 13th International LISA Symposium                                                                                                | <i>University of Auckland, remote</i>  |
| 02/20 | ANITA workshop and school 2020                                                                                                       | <i>UNSW, Canberra</i>                  |
| 01/20 | Gravitational Waves Group Meeting                                                                                                    | <i>Cardiff University</i>              |
| 01/20 | Astrophysics Seminar                                                                                                                 | <i>University of Birmingham</i>        |
| 12/19 | 2019 Stars in Melbourne                                                                                                              | <i>Monash University</i>               |
| 11/19 | 2019 OzGrav Annual Retreat                                                                                                           | <i>Lorne, Melbourne</i>                |
| 11/19 | OzGrav Data/Astro Telecon                                                                                                            | <i>Virtual</i>                         |

#### Softwares and code development

- Software development: COMPAS (rapid binary population synthesis), PHANTOM (smoothed particle hydrodynamics)
- User: MESA (stellar evolution), REBOUND (N-body integrator), Seven-League Hydro (finite-volume hydrodynamics), Phlegethon (finite-volume hydrodynamics), Athena++ (finite-volume hydrodynamics)
- Programming languages: Python, Fortran, C++, MATLAB